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Detecting faulty lithium-ion cells in large-scale parallel battery packs using current distributions

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Abstract

One of the main issues affecting the uptake of battery packs are safety concerns, particularly with respect to the fires caused by cell faults. Managing the risks from faults requires advances in battery management systems and an understanding of the dynamics of large packs. To address this issue, a machine learning classifier based upon a support vector machine was developed to detect cell faults within large packs using a limited number of current sensors. To train the classifier, a modelling framework for parallel connected packs was introduced and shown to generalise to Doyle-Fuller-Newman electrochemical models. The fault classification performance was found to be satisfactory, with an accuracy of 83% using current information from only 27% of the cells. These results highlight the potential of combining mathematical modelling and machine learning to improve battery management systems and deal with the complexities of large packs.

Applications such as transport electrification and the rise of grid storage as a means to accelerate the integration of renewables into the grid are leading to a rapid growth in the number of large lithium-ion (li-ion) battery packs.[1] However, as large packs become more widely adopted, concerns have been raised about their safety and the ability of the battery management systems (BMS) to detect faults that could eventually lead to fires.[2] Various faults can occur during the operation of a pack, mainly due to ageing and improper use,[3] and several instances of electric vehicle (EV) fires caused by battery pack faults have now been reported.[4] For example, the National Transportation Safety Board reported 17 lithium-ion battery fires on Tesla vehicles in the USA in 2018, out of a total of 350,000 vehicles.[5] Detecting these faults and isolating them before they become dangerous is therefore a growing concern for EV users. The challenge of fault detection is tackled by the pack's BMS,[6] for which pack safety is an often overlooked responsibility. However, the capabilities of the BMS varies with the pack's architectures (as in how the individual cells are connected in series and parallel), with each li-ion battery pack application generally having a different pack architecture as different energy and power requirements have to be met. Cells in EV battery packs are typically connected in parallel within individual modules, which are then connected up in series.[7] This arrangement is often referred to as the nPmS arrangement, where m modules of n parallel cells are connected in series. Within these parallel connected modules, both external faults and internal faults can occur.[8,9] While external faults are primarily caused by sensor faults and connection faults, the causes of internal faults include cells overcharge, over-discharge, short circuits, accelerated degradation, and thermal runaway,[10] a varied collection of issues that can be challenging to detect since each cell behaves as a black-box. While the formation of external faults can be sudden, internal faults often appear gradually, and thus their formation can be monitored and detected more effectively. This ability to detect and manage the formation of internal faults (even though doing so may be challenging using existing pack sensor technology) motivates the fault-detection algorithms of this work.

The problems associated with battery pack faults (including fires and deterioration in pack performance) have motivated research into improving the fault detection capabilities of the BMS. Cells within large-scale battery packs have intrinsic variability, caused by slight changes in their manufacturing and usage.[11] Within parallel connected packs, this cell-to-cell variability causes an uneven distribution of currents across the

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branches in the pack, resulting in the appearance of oscillations.[12–14] As the behaviour of these oscillations depends upon the cell-to-cell variability of the parallel pack, current measurements can then be used to encode information about the pack's health. Since current sensors are already widely deployed in packs for state-of-charge estimation (although, often, not all parallel branch currents are measured) and are relatively inexpensive, these sensors have the potential to be scalable, cost-effective, and accessible tools for diagnosing pack health and detecting faults. However, if pack diagnosis via current sensor information is to be a success, improvements in BMS algorithms are needed to effectively analyse the sensors' data streams.

Motivated by this need, this paper develops a fault detection method that uses the information encoded within the uneven pack current distributions of parallel connected packs to improve algorithm accuracy, using the following approach. First, a model of a battery pack is established, and the branch currents are simulated for a large parallel pack. Then, an internal cell fault detection algorithm based upon the support vector machine (SVM) is proposed using the simulated data. It should be noted that the use of the branch current distributions as a means of detecting cell faults has recently been demonstrated by Ding et al.[15] Their work estimated the current distributions based on terminal voltage, pack current and state-of-charge (SoC) using a Long Short-Term Memory (LSTM) network. They then compared these currents with the actual measured signals and used a generated residual signal allowing the detection of connection faults. The main differences with our work include the fact that Ding et al.[15] proposed a detection method for connection fault (external faults) whereas we study accelerated degradation faults (internal faults), and they focused on small packs (4p1s) whereas our results are based on large packs (74p1s). The modelling framework and analysis proposed in this paper also differ. In particular, the framework introduced in this work expresses the branch currents of the parallel pack explicitly in terms of the model states and the applied current. This formulation allows the differential algebraic equations (DAEs) of the pack model to be converted into a set of ordinary differential equations (ODEs), significantly simplifying the pack model equations. The obtained expressions for the branch currents are also shown to generalise to more complex battery models than the circuit models developed for the fault-detection algorithm, in particular to the Doyle-Fuller-Newman electrochemical model. As far as the authors are aware, these are the first simulations of parallel connected DFN models, a result which will enable detailed electrochemical simulations of li-ion battery packs in the future.

Three main methods have been used to detect battery faults in previous results: model-based methods, signal processing, and knowledge-based methods.[16] Model-based methods rely on battery models (mostly electrochemical models or equivalent circuit models) whose parameters are estimated using system identification techniques.[17,18] Measured and estimated signals are then compared and a residual signal is extracted to diagnose a potential fault.[19] Signal processing methods rely on large amounts of acquired battery data from which fault features are extracted (e.g. using entropy and wavelet transforms).[20,21] From these features, faults are detected once a given Z-score threshold has been exceeded by the measured data. Finally, the knowledge-based approach involves applying methods such as expert systems, fuzzy logic, and neural networks to diagnose battery faults.[22–24] Criteria describing the faulty state of the battery are established and thresholds defining the detection of a fault are determined.

The need for effective fault detection algorithms for battery pack BMS designs has motivated several studies on this topic. Tran et al.[25] proposed a model-based sensor fault diagnosis method and demonstrated how the equivalent circuit model (ECM) parameters were affected by cell degradation and by sensor faults, with the proposed method able to detect and isolate voltage and current sensor faults from the estimated cell degradation. Nuhic et al.[26] developed a data-driven approach for health diagnosis using a SVM, which was shown to effectively learn the degradation behaviour of li-ion cells. Yao et al.[27] also demonstrated the ability of a SVM classifier to identify a cell fault and give an indication of its severity. Finally, Sidhu et al.[28] proposed a model-based method to diagnose multiple faults, using extended Kalman filters to represent signature-fault models.

Building upon these earlier results, the primary contributions of this article are as follows. An equivalent circuit model of a parallel-connected battery pack is developed. This model is used to simulate single parallel modules (74p1s) within a Tesla Model S battery pack (74p96s) containing Panasonic NCR 18650B cells. An expression for the parallel pack branch currents is derived which converts the model's DAEs into ODEs. This expression for the currents is shown to hold for a broad class of cell-level models, in particular for parallel connected Doyle-Fuller-Newman electrochemical battery models. The current distributions from the 74p1s pack model simulations are then analysed and their response due to cell faults caused by accelerated degradation is investigated. A cell fault detection algorithm based on a support vector machine is then trained and applied. By demonstrating the potential of pack current data to diagnose faults in this way, these

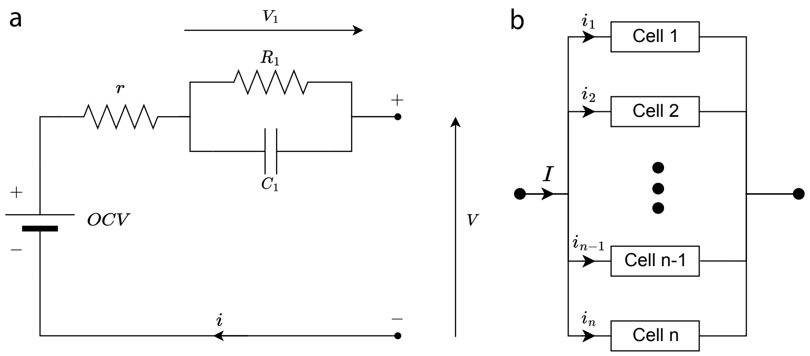


Figure 1: Equivalent circuit model (a) and module of n cells connected in parallel (b).

results highlight the potential of the proposed algorithm to improve pack safety and performance whilst being deployed on relatively inexpensive and readily available sensing hardware.

Results and Discussion

Model simulations

A model for parallel connections of li-ion cells that captures the impact of variations in the series resistance r and capacity Q on the response of the pack is developed. The key feature of this model is the way it resolves Kirchhoff's laws, which allows the DAEs of the pack model to be converted into ODEs, making it significantly simpler to solve. The class of cell-level models for which the following pack-level modelling results can be applied are those where voltage can be expressed as

$$V_k(t) = f(x_k(t)) + r_k i_k(t), \quad k = 1, 2, \ldots, n \tag{1}$$

where the index $k \in \{1,2,\ldots,n\}$ relates to the cell number in the pack, n is the total numbers of connected cells, $x_k(t)$ is the dynamic state of cell k which is associated with a time derivative in the model's equations, r_k is that cell's series resistance and $i_k(t)$ is the branch current (as in the current flowing into branch k of the parallel pack). Connecting these cell-level models in parallel means that Kirchhoff's laws have to be satisfied to compute the branch currents $i_k(t)$

$$f(x_j(t)) + r_j i_j(t) = f(x_k(t)) + r_k i_k(t), \quad j,k \in \{1,2,\ldots,n\}, \tag{2a}$$

$$\sum_{k=1}^{n} i_k(t) = I(t). \tag{2b}$$

Combining Kirchhoff's laws of Equation 2 with the cell dynamics means that parallel connected pack models are described by a set of DAEs.[29] However, using the approach outlined in the Methods section, the algebraic equations of Kirchhoff's laws from Equation 2 can be resolved to give

$$i_1(t) = \left(r_1 \sum_{k=1}^{n} \frac{1}{r_k} \right)^{-1} \left(\sum_{k=2}^{n} \frac{\Delta f_{k1}(t)}{r_k} + I(t) \right), \tag{3a}$$

$$i_k(t) = \frac{1}{r_k} \left(r_1 i_1(t) - \Delta f_{k1}(t) \right), \quad k = 2, 3, n, \tag{3b}$$

where $\Delta f_{jk}(t) = f(x_j(t)) - f(x_k(t))$. With this expression, the pack's branch currents can be written as a function of the pack's states $x_k(t)$ and the applied pack current $I(t)$. Doing so converts the parallel pack models dynamics from a systems of DAEs to ODEs, since the algebraic equations of Equation 2 are now expressed in terms of the model's states and the applied current.

Resolving Kirchhoff's laws using Equation 3 means that the parallel pack model does not have to numerically solve Equation 2 at each time step of the simulation to compute the branch currents $i_k(t)$. Instead, these currents are defined by Equation 3 and are expressed in terms of the cell models' states $x_k(t)$ and the applied pack current $I(t)$, allowing the pack model dynamics to be solved as an ODE rather than a DAE.[29] Turning the DAE parallel pack model into an ODE brings insight into the distribution of currents between the branches in the pack, in contrast to the numerical approach where the computational solution

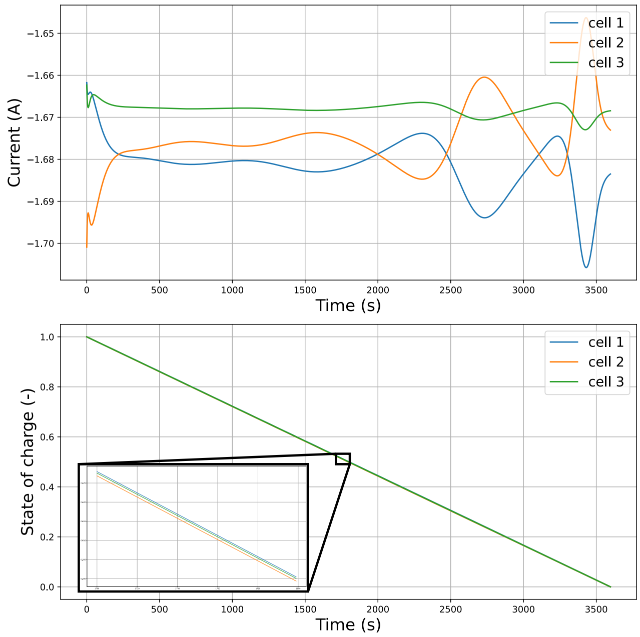


Figure 2: Currents and states-of-charges simulated by the model for 3 parallel-connected cells during a 1C discharge.

only gives a limited insight into the mechanisms by which the pack's currents re-distribute themselves. The numerical solution for computing the parallel pack branch currents is, however, widely used, in both modelling studies,[30–33] estimator designs[34–37] and modelling results based upon the PyBAMM[38] open-source software.[39] The extent to which this computational solution has been used indicates the applicability of the analytic solution of Equation 3 to understand these nonlinear parallel pack dynamics. Moreover, the proposed approach overcomes some of the restrictive assumptions of existing analytical solutions, such as the *OCV* being affine[40,41] and the cell-level model simply being a voltage source and a series resistor.[42,43] Instead, with Equation 3, the voltage is allowed to be of the form of Equation 1, as in the sum of a linear series resistance term and a nonlinear function of the states. Compared to the existing derivation for the branch currents from Drummond and Duncan,[44] the method presented here is simpler since it does not involve computing a matrix inverse. Instead, in this work, the structure of the equations defining Kirchhoff's laws are exploited to solve Equation 2.

The cell-level used to design the fault-detection algorithm for the parallel pack branch currents was the equivalent circuit model described in the Methods section. The different cell parameters were assumed to be normally distributed from cell-to-cell. The means of these parameter distributions are given in Table 2 and the standard deviations are given in Table 3, with these values extrapolated from the results of Schneider et al.[45] Since the values observed in[45] were based on different cells (Samsung INR18650-25R cells) from those considered here (Panasonic NCR 18650B cells), the σ/μ ratios from[45] were first computed and then related to the mean values from Table 2 to derive the standard deviation values for the NCR 18650B cell considered here. The standard deviations computed using this method are stated in Table 3.

To analyse the behaviour of this model, a 1C discharge of three cells connected in parallel was considered. The simulated currents and state-of-charge values are plotted in Figure 2, and it is noted that the current distributions for these 3 cells are similar in shape to those measured experimentally by Chang et al.[46] In both cases, the characteristic current oscillations of parallel connected packs are observed. It is the properties of these oscillations that are examined in this work for the purpose of fault detection.

The architecture of a Tesla Model S battery pack was used as a reference for a large parallel-connected battery pack. This pack has a 74p96s configuration, but a single module (74p1s configuration) was simulated here. Since all the modules are subjected to the same current pack, they all behave similarly, which explains the focus of these results on the simulation of a single module. The 74 cells of the module were assigned parameters according to the normal distribution defined previously (Table 2 and Table 3), and the ODEs of the pack model were solved to obtain the terminal voltage, state-of-charge, and current flowing through each cell in parallel during a full discharge. The results of this pack level simulation are shown in Figure 3a. The observed current oscillations typically have an amplitude of a few hundredths of an ampere, and are therefore detectable with shunt current sensors, which typically have tolerances ranging from 0.1% to 1% and can be connected in series with the cells.[14] The highest current deviations from the mean are located at the beginning and the end of the discharge. At the start of the simulation, the immediate distribution in the currents across the pack is due to the differences in series resistances from cell-to-cell: different values of r imply that different currents must flow through the cells if they are to have the same terminal voltage. Because of this initial current distribution, the cells discharge at different rates. The values of the cell OCVs will then also vary from cell-to-cell, since the OCV is a function of the state-of-charge, a feature that causes the currents within the pack to rebalance themselves to enforce Kirchhoff's voltage laws. This rebalancing mechanism is the reason why the oscillations are more visible at the end of the discharge where the slope of the OCV is steep. It should also be noted from Figure 3a that the SoCs varies slightly between the cells and that the terminal voltage is the same throughout the parallel pack, which is consistent with the Kirchhoff law equations of Equation 2.

It should be recalled that the cells modelled during this study were considered to be fresh and from the same batch, implying little cell-to-cell variability (except for the faulty cells used for the fault detection algorithm which were deliberately selected to have a high variation so as to represent a fault). For comparison, a simulation of the same pack but with the fresh cells replaced with aged ones was run, as shown in Figure 3b. To characterise the aged cells, the value of the standard deviations of the parameters was multiplied by 5 with respect to the values chosen previously (see Table 3). The current variations follow a similar shape to the fresh pack simulation of Figure 3a, with one significant difference; the amplitude of the variations is one order of magnitude higher. This increased variation in the currents is responsible for the significantly higher SoC deviations of the aged pack when compared to the healthy pack results in Figure 3a, which are amplified further by the increased variation in the capacitance of each cell. These capacitance differences also cause a larger shift in the current oscillations over time; the current deviations of the cells with the highest capacitances have a slight delay compared to others, which is particularly noticeable at around 2800 seconds in the simulation. The simulations suggest that a degradation fault would be more easily identifiable in aged packs compared to fresh ones, as the current oscillations would be greater. In the remainder of this paper, the focus will be on the design of fault detection algorithms for the fresh pack described above.

The general form of the cell-level voltage equations of Equation 1 allows the equation for the branch currents given in Equation 3 to be applied to a broader class of models than just the circuit model discussed above. In particular, it can be applied to DFN electrochemical li-ion battery models[47] with double layer effects included[48,49]- a benchmark electrochemical model for li-ion batteries. Electrochemical battery models are, generally, more complex than circuit models, and this added complexity has introduced challenges when using them within pack models, as observed in, for example, the recent work of Reniers and Howey.[50] In that work, a pack-level model for a 1 MWh grid battery system containing 18,900 cells and cycled for 10 years was develloped. The cell-level model of that study was the single particle model (SPM), and, even though the SPM is one of the simplest forms of battery electrochemical models, the analysis of that work highlighted the difficulty in resolving Kirchhoff's laws for parallel connections of SPMs. Specifically, due to the SPM's nonlinear series resistance, the parallel Kirchhoff laws had to be resolved using an approach based upon a PID controller, which introduced some errors.

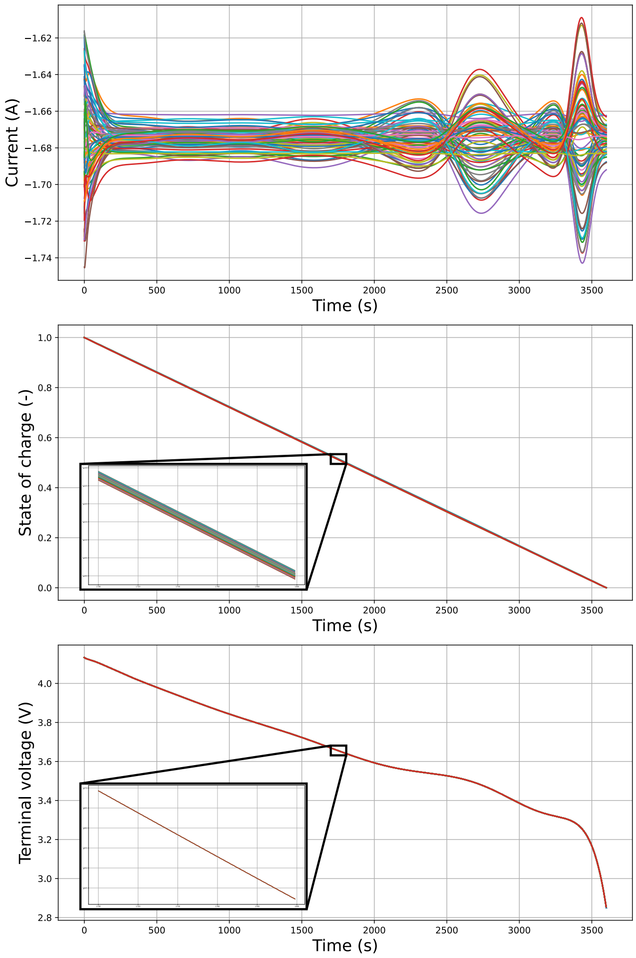
By contrast, it is shown in the Supplementary Information (with SI1 giving the mathematical analysis and SI2 defining the DFN model's variables and parameters) that resolving parallel connections of DFN models is, in fact, more intuitive and, arguably, easier to implement, than for the SPM. Even though the DFN model is more complex than the SPM (it is in fact a generalisation), this result shows how computing the branch currents is actually easier. As detailed in SI1, this result is obtained by expressing the DFN model voltage in the form of Equation 1 with

$$r = R_{\text{ctc}} + \int_{\Omega_\text{n}} \frac{1}{\sigma_\text{s} + \kappa_\text{e}}\, dx + \int_{\Omega_\text{p}} \frac{1}{\sigma_\text{s} + \kappa_\text{e}}\, dx + \int_{\Omega_\text{s}} \frac{1}{\kappa_\text{e}}\, dx,$$

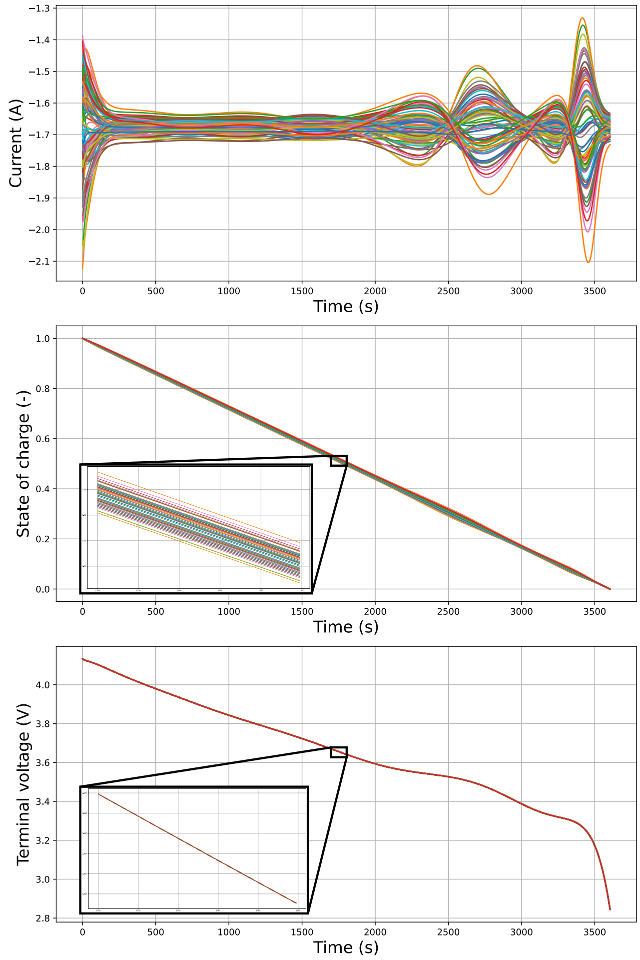
and

$$\begin{aligned} f(x_k(t)) = \phi_{\text{dl}}(L,t) - \phi_{\text{dl}}(0,t) &+ \int_{\Omega_n} \frac{\omega\kappa_\text{e}}{\sigma_\text{s}+\kappa_\text{e}} \frac{\partial \ln(c_\text{e}(x,t))}{\partial x} - \frac{\sigma_\text{s}}{\sigma_\text{s}+\kappa_\text{e}} \frac{\partial \phi_{\text{dl}}(x,t)}{\partial x}\, dx \\ &+ \int_{\Omega_\text{p}} \frac{\omega\kappa_\text{e}}{\sigma_\text{s}+\kappa_\text{e}} \frac{\partial \ln(c_\text{e}(x,t))}{\partial x} - \frac{\sigma_\text{s}}{\sigma_\text{s}+\kappa_\text{e}} \frac{\partial \phi_{\text{dl}}(x,t)}{\partial x}\, dx + \int_{\Omega_\text{s}} \omega \frac{\partial \ln(c_\text{e}(x,t))}{\partial x}\, dx. \end{aligned}$$

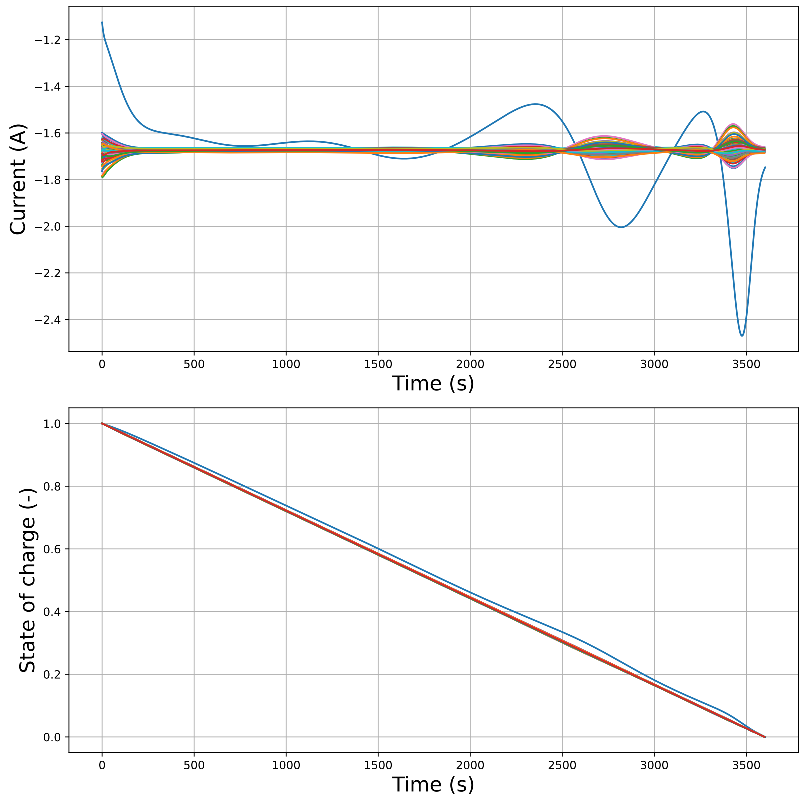
Crucially, by including double-layer dynamics in the DFN model, the potential $\phi_{\text{dl}}(x,t)$ becomes a model state as the double-layer dynamics give it a time derivative. The parallel branch currents can then be computed directly using Equation 3, eliminating the need to solve the algebraic equations of Kirchhoff's laws when



(a) Currents, state-of-charges and voltages simulated by the model for 74 parallel-connected cells (fresh Tesla Model S module) during a 1C discharge.



(b) Currents, states of charge and voltage simulated by the model for 74 parallel-connected cells (aged Tesla Model S module) during a 1C discharge.



(c) Simulated currents and state-of-charges for 74 parallel-connected cells with one faulty cell during a 1C discharge.

Figure 3: Currents, state-of-charges and voltages of several battery packs simulated by the model.

using DFN models. Thus, even though the DFN is a more complex electrochemical model than the SPM, computing the branch currents for parallel connections is intuitive and can be readily implemented, with the branch currents expressed directly as a function of the model's states and the applied current.

To verify that Kirchhoff's laws had been satisfied with this DFN pack model, simulations of two parallel connected DFN electrochemical models for lithium-ion batteries were conducted. The parameters of these simulations are given in SI2. The model is solved numerically by discretising the model's PDEs using second order central differences to approximate the spatial derivatives, along with averaging of the diffusivity and conductivity functions at the control volume faces and solved using "`ode15s`" within `MATLAB`® 2021b.[51] The parameters for each cell in this pack model are given in Table S1 and Table S2. A constant current discharge of the pack was modelled using a discharge current density of $I(t) = 129.55\ A/m^2$. It was assumed that the contact resistance between cells one and two in the simulated parallel pack differed with $R_{\text{ctc}} = 5.7 \times 10^{-4}$ for cell one and $R_{\text{ctc}} = 7.7 \times 10^{-4}$ for cell two- this setup was referred to as case 1. As well as differences between the cells' contact resistances, two other cases were simulated. Case two additionally had the anode reaction rate coefficient in cell two being three times that of cell one while case three had the anode particle radius of cell two being three times greater in cell two than in cell one.

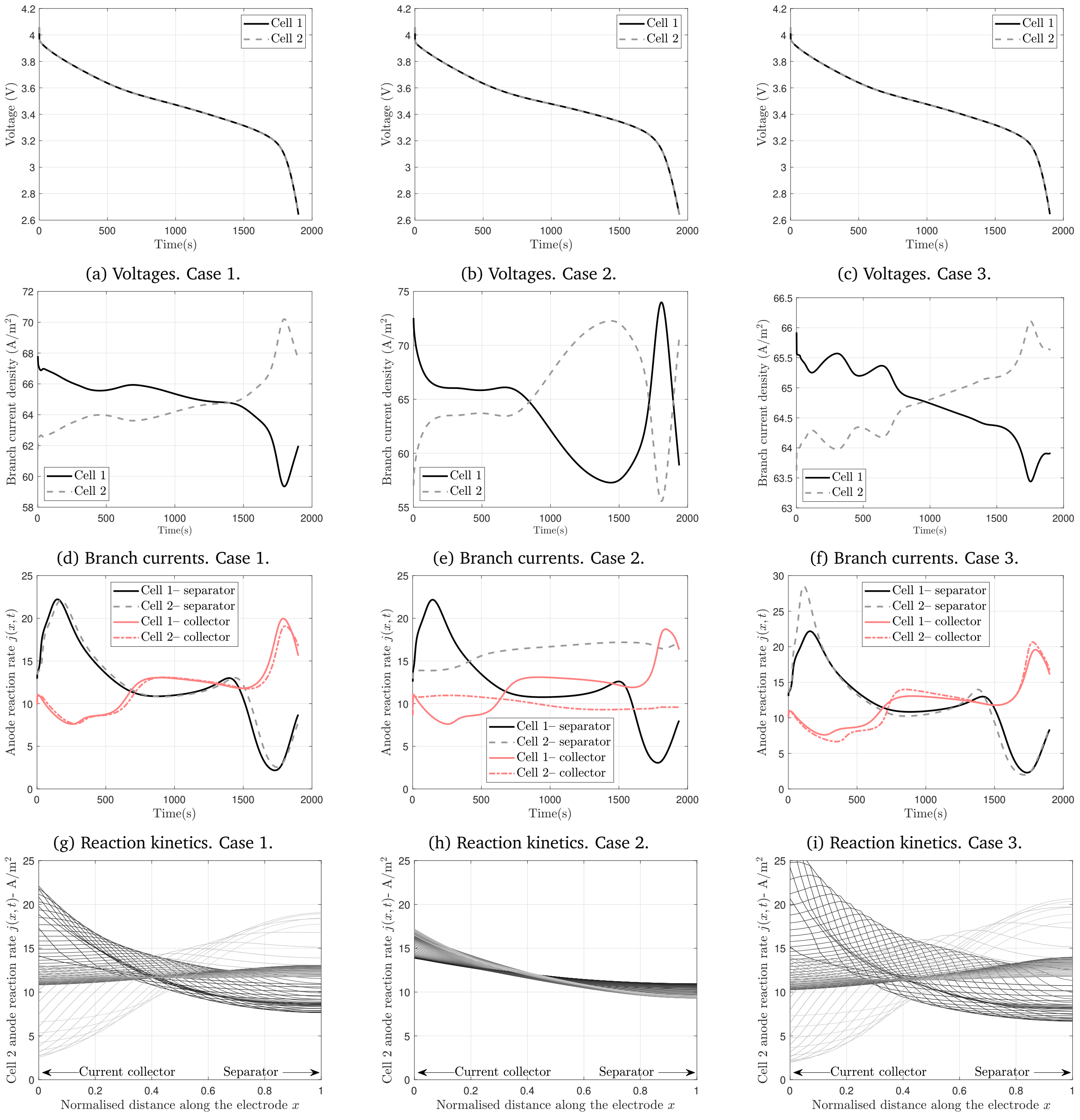
The simulations results for two DFN models connected in parallel for the three cases discussed above are shown in Figure 4. In this figure, the first row describes the pack voltages, the second row is the branch currents, the third row is the evolution of the cathode reaction rate kinetics at both the current collector and the separator boundaries, and the fourth rows corresponds to the evolution of the spatial distribution of the reaction kinetics through the thickness of the anode. Each line in the fourth row figures correspond to a snapshot of $j(x,t)$ in the anode taken every 25s during the simulation, with the darker lines corresponding to the start of the simulation and the lighter lines corresponding to the end. The three columns in the figure corresponds to the three different cases of the cell parameters being simulated.

From the current and voltage plots of Figure 4, it can be observed that Kirchhoff's laws are satisfied; both cell voltages are equal and the current splits between the two cells and constantly rebalances itself during the discharge. The three different cases were found to lead to three different responses, highlighting the degree to which manufacturing variability of the cells (and hence, the variation in their electrochemical parameters) can affect pack performance. The final two rows in the figure illustrate how variations in the cell parameters can lead to variations in the electrochemical response. These variations may impact the pack's response in the long-term, noticeably on the cell degradation rates.

The generalisation to parallel connected DFN models discussed above highlights how the modelling framework of Equation 2 can be applied to more complex models than the simple circuit models previously considered. With this approach, detailed simulations of the pack's electrochemical response can be readily simulated. However, due to the significant computational complexity and lack of effective parameter estimation methods for DFN models, it was decided to base the fault detection algorithm of this work on the parameterised circuit models for the 74p96s Tesla Model S modules. Once the computational and parameterisation issues of DFN models are resolved, the expression for the branch currents stated in Equation 3 can then be used for pack-level problems involving DFN models, such as designing advanced fault-tolerant algorithms. The additional information on the electrochemical state of the pack captured by these models is expected to improve the performance of battery management system algorithms when compared to those based upon circuit models, and will be explored in the future.

Fault detection

To implement the cell fault detection algorithm, a characterisation of a cell fault must first be introduced. Here, the focus is on modelling faults caused by accelerated degradation. Amongst other phenomena, as a cell ages, its resistance increases, especially at the end-of-life.[52] Within a pack, some cells will reach this stage sooner than others, leading to potential accelerated degradation faults. For this reason, the resistances of the cells within a pack can be regarded as useful indicators of cell faults. We, therefore, considered the detection of faults caused by an abnormal increase in the ohmic resistance of a cell due to accelerated degradation.[53] According to the cell model described in Figure 1, a cell whose series resistance r follows a normal distribution based upon the values given in Table 2 and Table 3 will be defined as healthy. A faulty cell will be defined by an abnormally high resistance r (at least 5 standard deviations above the mean). Experimental studies such as[54] have shown that series resistances of cells within packs can vary by several dozen percent during ageing. For this reason, the following analysis will consider the case when the faulty cell has an increased resistance ranging from 10% to 100%.



(a) Voltages. Case 1. (b) Voltages. Case 2. (c) Voltages. Case 3.

(d) Branch currents. Case 1. (e) Branch currents. Case 2. (f) Branch currents. Case 3.

(g) Reaction kinetics. Case 1. (h) Reaction kinetics. Case 2. (i) Reaction kinetics. Case 3.

(j) Cell 2 $j(x,t)$. Case 1. (k) Cell 2 $j(x,t)$. Case 2. (l) Cell 2 $j(x,t)$. Case 3.

Figure 4: Discharge simulation results of two parallel connected DFN models using the parameter values of Table S1 and Table S2. Case 1 corresponds to the nominal setup, with $R_{\text{ctc}} = 5.7 \times 10^{-4}$ for cell 1 and $R_{\text{ctc}} = 7.7 \times 10^{-4}$ for cell 2. Case two additionally has an anode reaction rate coefficient k three times greater in cell 2 than cell 1. Case three has an anode active particle radius three times greater in cell two than cell one and the fourth row is plotted every 25 seconds, with the dark lines corresponding to the start of the simulation and the lights ones to the end.

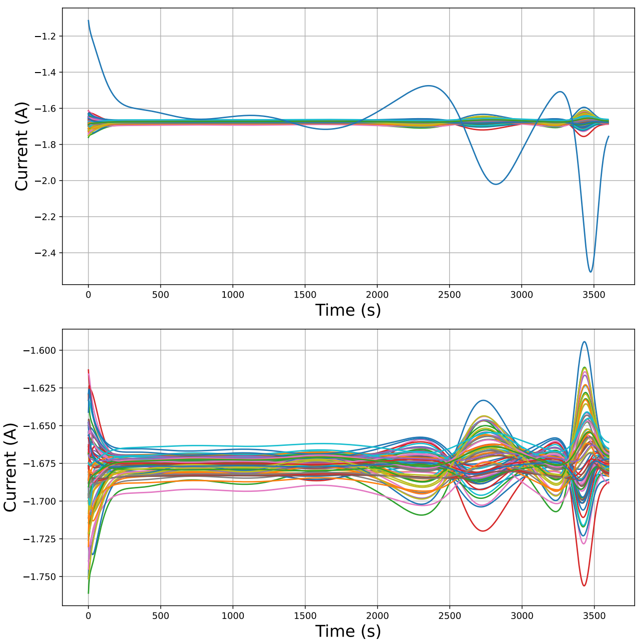


Figure 5: Comparison between the current distributions of a faulty pack (top) and the same pack without the faulty cell current (bottom) during a 1C discharge.

To evaluate the impact of such cell faults on the current distributions across parallel connected packs, simulations of both a healthy pack (see Figure 3a) and one with a fault were then carried out. The pack containing the fault was similar in all aspects with the healthy pack except for the fact that one of the cells had an increased series resistance, $r = 1.5 \times \mu_r = 28.5\,\mathrm{m\Omega}$. The results obtained are shown in Figure 3c. There is a clear difference compared to the healthy pack seen in Figure 3a: the faulty cell exhibits a discharge current at $t = 0$ with a low absolute value, due to its resistance fault. Consequently, the current variations experienced by this cell are extreme in amplitude. However, in general, the currents flowing through the non-faulty cells in this aged pack all behave similarly to those obtained for a healthy pack. This similarity between the currents in fresh and aged packs introduces a challenge for fault detection when not all the currents are being monitored.

A fault detection algorithm based upon a SVM classifier was trained on the dataset to perform the binary classification between healthy and faulty parallel packs based upon the simulated current distributions, with the details of this algorithm given in the Methods section. Features for the developed SVM fault detection algorithm were extracted from the dataset samples in order to discriminate the healthy packs from the faulty packs.

The performance of the classifier was first verified for detecting faulty packs when given all the simulated currents as inputs. Using simple features based on the deviation of the currents from the mean, it was expected that faulty packs could be detected, as they presented an abnormal current signal (see fault detection section). Once trained, the classifier was able to classify the 200 samples of the training set without making any classification errors, thus achieving a 100% accuracy score. The algorithm thus performed successfully when all the currents in the pack were being monitored.

However, in practice, only a limited number of branch currents are typically measured with large parallel connected packs, due to the increased costs from the sensors and because they add to the system complexity. Thus, to make the proposed algorithm more applicable, the fault detection algorithm for the sparse sensing problem, as in when only a fraction of the number of branch currents were measured, was then considered. With sparse sensing, the fault detection becomes significantly harder, as it is challenging to infer the impact of the faulty cell on the rest of the pack when only measuring the healthy cells' currents. For this analysis, it was assumed that only N_s current sensors were deployed in the pack and it was ensured that the current from the faulty cell was not measured by these sensors. Features for the developed SVM fault detection algorithm were extracted from the dataset samples in order to discriminate the healthy packs from the faulty packs. The extracted features described in the Methods section were then used and an ablation study was performed to characterise their impact in the classification, with it being observed that each feature contributed to the accuracy of the classifier. After empirically testing and selecting different features, it was noticed that the most effective descriptors were formed from the local minima and maxima of the current signals. The

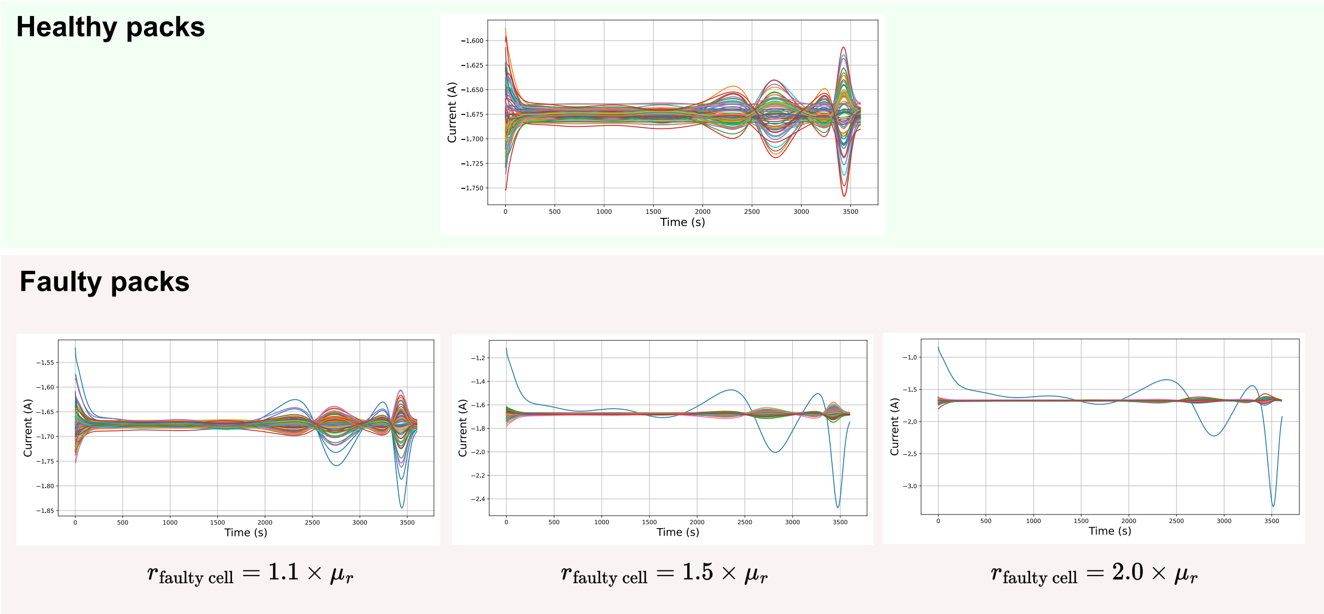


Figure 6: Simulations of the Tesla Model S battery pack module when all the cells are healthy (top) and when one of the cells is faulty (bottom). The value of the faulty cell's resistance ($r_{\text{faulty cell}}$ which is defined as a multiple of the pack's mean resistance μ_r) significantly impacts the pack's current distribution, and hence the performance of the fault classification algorithm.

classification was performed for a range of sensors N_s in the parallel pack. The classifier was first trained and tested for $N_s = 73$, i.e. all cell currents given as input except the faulty cell current. The confusion matrix of the corresponding prediction results is given in Table 1a. The same procedure was then applied for $N_s = 20$, i.e. sensors placed on only a subset of the 74 cells with the results are given in Table 1b.

However, in practice, these measurements may not be available as current sensors are often not added to every branch of the parallel connected pack. In fact, often only a few sensors are used since these sensors add to costs and require additional compute power from the BMS to process the data. For these reasons, a fault detection algorithm must be robust in the sense that it should still function even when the current of the parallel branch is not measured. The results of the following section address this issue.

The features were then trained on the dataset generated by this model, whose structure is shown in Figure 6. The profile of the current distributions simulated within the dataset can be seen in this figure, which shows, with the exception of the case $r_{faulty\ cell} = 1.1 \times \mu_r$, that faulty packs can be easily distinguished from healthy packs with measurements of the faulty cell current. This again highlights the increased in difficulty in designing the fault detection algorithm for the sparse sensing problem. After calculating the features for all samples in the training and testing set, their histograms were then analysed. These histograms were calculated for $N_s = 73$ (i.e. the current signal from the faulty cell was removed in faulty packs, while a random current was removed for healthy packs). The values taken by the features for the healthy and faulty packs were separated, with the resulting histograms compared in Figure 7. It is noted that the observed distributions for these features differ between healthy and faulty packs, except for f_1. Although from Figure 3a and the fault detection algorithm results, it would appear that the current distributions between the cells in a healthy pack and the healthy cells in a faulty pack are similar, the histograms reveal that the extrema reached by the currents, which form the basis of the six identified features f_1 to f_6, allow one to partially discriminate the two classes. It is this discrimination that allows the SVM algorithm to perform the fault detection, even when only limited sensor information was used. After empirically testing and selecting different features, it was also noticed that the most effective descriptors were formed from the local minima and maxima of the current signals. Finally, it is noted that as the number of removed current sensors increased, then the distributions of the histograms of the two pack classes (healthy and faulty) appeared to overlap, resulting in a reduced discrimination ability of the classifier.

Finally, the classifier was trained and tested for a larger number of N_s values, between 2 and 72, and the accuracy of the model was evaluated for each of these trials. The results of these trials, characterising the performance of the proposed detection method, and constituting the main result of this paper, are shown in Figure 8. This figure compares the accuracy of the proposed faulty pack detection algorithm as a function of the number of measured current branches N_s. A plateau can be seen in the prediction accuracy of the figure at around 20 sensors out of 74, with the plateau continuing to around 50 sensors. This plateau suggests

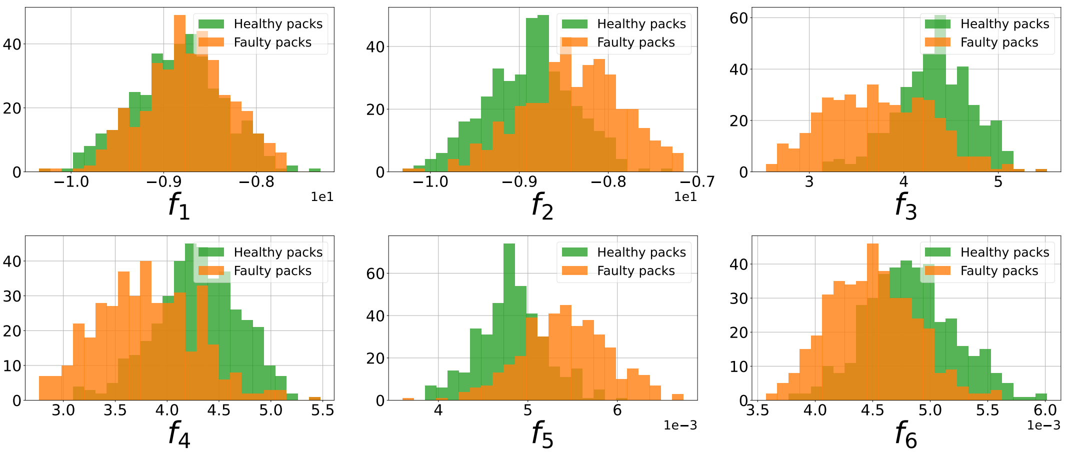


Figure 7: Histograms of the feature distributions for the healthy and faulty packs of the dataset.

		Prediction		
		Faulty pack	Healthy pack	Total
Truth	Faulty pack	81	17	98
	Healthy pack	5	97	102
	Total	86	114	200

(a) Confusion matrix of the SVM classifier with $N_s = 73$ (as in when all but the faulty parallel branch currents are measured).

		Prediction		
		Faulty pack	Healthy pack	Total
Truth	Faulty pack	76	27	103
	Healthy pack	7	90	97
	Total	83	117	200

(b) Confusion matrix of the SVM classifier for $N_s = 20$.

Table 1: Confusion matrices of the SVM classifier for two sensor configurations.

that the optimal detection configuration for the module of the Tesla Model S, in terms of precision and sensing cost, is around 20 sensors. As the number of deployed sensors decreased, the accuracy score also decreased: for 7 sensors (10% of the cells in the pack), an accuracy of 76% was obtained, and for 2 sensors only, this number dropped to 67%. Table 1b details the predictions made by the algorithm at this point and it is noted that the results contain few false positives, and a larger quantity of false negatives. Since an accelerated cell degradation fault in a pack is a relatively infrequent phenomenon, it is important that there are few false positives. However, false negatives can constitute a problem in terms of detection. It can be seen that this is the main difference with the $N_s = 73$ case discussed in Table 1a, where there are fewer false negatives. Considering the plot shown in Figure 8, and the statistical nature of the extracted features used by the detection method, it appears that this method is applicable for a parallel pack made of a large number of N_c cells, such as the Tesla Model S battery pack modelled here. However, for pack configurations where only a few cells are connected in parallel in each module, a method that uses only a small fraction of the total number of cells in the module may prove difficult to implement.

Several of the model parameters were found to strongly influence the dynamic response of the current distributions across the parallel connected pack and, consequently, the performance of the detection algorithm. For example, it was observed that the points where the oscillations in the current distributions were largest, during both discharging and charging, correlated with the points where the slope of the OCV curves were steepest. The shape of the OCV curve (which is different for each cell chemistry) therefore strongly impacts the ability of the algorithm to detect cell faults. Additionally, it is noted that the the ageing of the pack will influence the detection algorithm. As the pack ages, the variation between the cell parameters will grow, which in turn will lead to greater current deviations and potentially improve the accuracy of the detection algorithm (since the current data will be richer with more variations, making it easier to detect faults). A fresh pack was assumed in this study, so these detection results again could be said to provide a conservative baseline. It should also be noted that the obtained accuracy of 82.8% for $N_s = 20$ (i.e. sensors placed on about 27% of the cells in the pack) assumes that no sensor is ever placed on the faulty cell. Since this

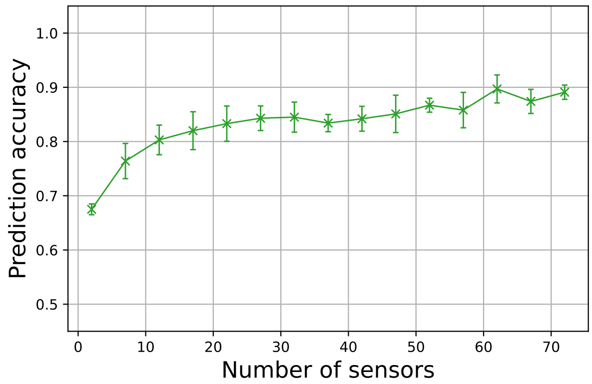


Figure 8: Accuracy score of the fault detection classifier (SVM) as a function of N_s (the number of current sensors within the 74p1s parallel pack).

should happen in this configuration in 27% of cases and the fault is systematically detected in this case, we can assume a higher accuracy in the general case. The accuracy scores of Figure 8 can therefore again be considered conservative baselines.

Conclusions

A method to detect cell faults in large parallel connected battery packs which combines model-based prediction with machine learning classification was presented. An equivalent circuit model of a parallel battery pack was developed and used to simulate the behaviour of an EV battery pack, specifically a Tesla Model S module. Current distributions within the parallel connected pack were studied in detail and an expression for the branch currents as a function of the model's states and the pack's applied current was derived. This expression for the parallel pack branch currents was shown to generalise to more complex battery models, including the Doyle-Fuller-Newman electrochemical model. The branch currents of the parallel connected were used within an SVM-based algorithm to detect faulty cells in the pack to improve its safety. The proposed fault-detection algorithm was shown to perform well, achieving an accuracy of 83% when using only 20 current sensors per module of 74 cells in parallel. These results demonstrate the potential of detecting cell degradation faults with sparse sensing on large-scale parallel battery packs. By combining algorithms with current sensing data in this way, these results can be used to improve the safety of a battery pack.

There is significant scope to develop the proposed results in future work. Notably, the authors are interested in deploying the proposed algorithm to monitor experimental battery packs. Since the proposed modelling framework and algorithm structure are relatively simple, it is expected that the proposed approach could be readily deployed to experimental setups. The authors are also interested in comparing the efficacy of the proposed SVM algorithm with approaches based upon deep learning, such as convolutional neural networks (CNNs). As these algorithms have delivered strong results in other classification tasks, it is expected that they may also perform well for the problem considered here of detecting faults in large parallel connected packs. However, neural networks have limitations which may affect their performance in practice. Notably, their lack of robustness and explainability may impact the extent to which engineers trust them for monitoring large and expensive packs, where safety issues need to be understood and failures explained. Preliminary results by the authors exploring the use of CNNs for this problem have, so far, not delivered satisfactory results, but further gains are expected in the future after further fine-tuning of the CNN.

Methods

Cell model

The pack model is based on the equivalent circuit model of a single cell and the connection of multiple cells in parallel. The cell model used is the Thevenin model shown in Figure 1 which has been widely used due to its simplicity combined with its high accuracy.[55] In this model, the dynamics for each cell $k \in \{1, 2, \dots, n\}$ are characterised by its series resistance r_k, RC-pair resistance $R_{1,k}$, RC-pair capacitance $C_{1,k}$ and capacity

Parameter	Mean Value	Unit	Reference
r	19	mΩ	57
R_1	1.7	mΩ	57
C_1	5598	F	57
Q	3350	mAh	58

Table 2: Mean values of the cell model parameters.

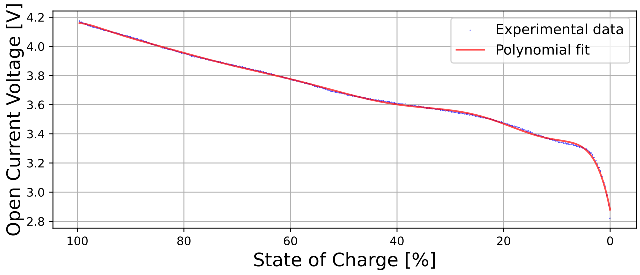


Figure 9: Comparison between the digitised OCV curve from[56] (blue) and its polynomial interpolation (red).

Q_k. The subscript k is dropped from these parameter labels when referring to them in a general context, for example in Table 2 where their mean values are stated. The battery pack of a Tesla Model S was modelled and so the modelled pack's cells were chosen to be Panasonic NCR 18650B cells. The corresponding model parameters are given in Table 2 whilst the OCV curve was obtained from experimental data[56] and fitted using a polynomial approximation (see Figure 9). The branch current $i_k(t)$ flowing through cell k was considered to be positive when charging and negative when discharging.

Within the pack model, the dynamics for cell $k \in \{1, 2, \dots, n\}$ were described by the simple equivalent circuit model

$$\begin{bmatrix} \dot{z}_k(t) \\ \dot{V}_{1,k}(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{R_k C_k} \end{bmatrix} \begin{bmatrix} z_k(t) \\ V_{1,k}(t) \end{bmatrix} + \begin{bmatrix} 1/Q_k \\ 1/C_{1,k} \end{bmatrix} i_k(t), \tag{4a}$$

$$V(t) = OCV(z_k(t)) + V_{1,k}(t) + r_k i(t), \tag{4b}$$

where, for cell $k \in \{1, 2, \dots, n\}$, $V_{1,k}(t)$ is the voltage drop across the RC-pair of Figure 1, $z_k(t)$ is the state-of-charge, $V(t)$ is the voltage and $OCV(z(t))$ is the open circuit voltage.

Parallel pack model: Conversion of DAEs into ODEs

The parallel connection of n cells, as described in Figure 1, was then modelled in accordance with Kirchhoff's laws. For parallel connections, these laws impose that the voltages $V(t)$ across each of the cells are the same and that the sum of the cell branch currents must equal the total current applied to the pack.

Defining $x_k(t)$ as the state vector of cell k, which for the circuit model is $x_k(t) = [z_k(t), V_{1,k}(t)]^\top$, and $i_k(t)$ as the current flowing through cell k in response to the pack current $I(t)$, then, for $j, k \in \{1, 2, \dots, n\}$, Kirchhoff's laws for parallel connected packs can be written as

$$OCV(z_j(t)) + V_{1,j}(t) + r_j i_j(t) = OCV(z_k(t)) + V_{1,k}(t) + r_k i_k(t), \tag{5a}$$

$$\sum_{k=1}^{n} i_k(t) = I(t). \tag{5b}$$

Combining Equation 4 and Equation 2 gives the system of DAEs of the parallel pack model. The use of DAE models is, however, often undesired because they can be significantly more challenging to simulate and analyse than those described by ODEs. In the following analysis, a method to translate this DAE model of the parallel pack into an ODE one is described. The first step of this conversion is to express Kirchhoff's laws as

$$V(t) = r_k i_k(t) + V_{1,k}(t) + OCV(z_k(t)) \tag{6a}$$

$$= r_k i_k(t) + f(x_k(t)), \forall k \in \{1, 2, \dots, n\}, \tag{6b}$$

where $f(x_k(t))$ is a function of cell k's state-space. Kirchoff's laws can then alternatively be written as

$$r_1 i_1(t) - r_2 i_2(t) = \Delta f_{21}(t), \tag{7a}$$

$$\vdots$$

$$r_1 i_1(t) - r_n i_n(t) = \Delta f_{n1}(t), \tag{7b}$$

$$i_1(t) + i_2(t) + \cdots + i_n(t) = I(t), \tag{7c}$$

where $\Delta f_{jk}(t) = f(x_j(t)) - f(x_k(t))$. With this formulation, each of the branch currents $i_k(t)$ can be expressed in terms of the first one, as in

$$i_2(t) = \frac{1}{r_2}\left(r_1 i_1(t) - \Delta f_{21}(t)\right), \tag{8a}$$

$$\vdots$$

$$i_n(t) = \frac{1}{r_n}\left(r_1 i_1(t) - \Delta f_{n1}(t)\right). \tag{8b}$$

Substituting these expressions back into Equation 7c gives

$$I(t) = i_1(t) + i_2(t) + \cdots + i_n(t), \tag{9a}$$

$$= i_1(t) + \sum_{k=2}^{n} \frac{1}{r_k}\left(r_1 i_1(t) - \Delta f_{k1}(t)\right), \tag{9b}$$

$$= \left(1 + r_1 \sum_{k=2}^{n} \frac{1}{r_k}\right) i_1(t) - \sum_{k=2}^{n} \frac{\Delta f_{k1}(t)}{r_k}. \tag{9c}$$

The first branch current $i_1(t)$ can then be written exclusively in terms of the model's state and the applied current via

$$i_1(t) = \left(r_1 \sum_{k=1}^{n} \frac{1}{r_k}\right)^{-1} \left(\sum_{k=2}^{n} \frac{\Delta f_{k1}(t)}{r_k} + I(t)\right). \tag{10}$$

Parameter	μ/σ ratio	Standard deviation σ	Unit
r	2.09 %	0.40	mΩ
R_1	1.64 %	0.028	mΩ
C_1	7.12 %	399	F
Q	0.28 %	9.4	mAh

Table 3: Standard deviations of the cell model parameters.

Using Equation 3, the other cell currents $i_2, \ldots, i_n$ can then also be written in terms of the model states and the applied current. This process allows the parallel pack model to be described by an ODE instead of a DAE, which greatly simplifies the analysis.[29] The ODE model equations were then solved numerically using the `integrate.odeint` solver from the `scipy` package in `Python`.

Data processing pipeline

To implement the faulty cell detection method, the data processing pipeline represented in Figure 10 is introduced. The first step of this pipeline is to simulate a large quantity of battery packs discharges (both healthy and faulty) using the pack model developed around Equation 2 to generate a substantial number of current distributions. The second step consists in pre-processing the data to better represent its acquisition by the current sensors used in practice, and to separate the generated data into a training set and a testing set. In the third step, heuristic features were built to facilitate the classification.

Each pack was characterised by a set of 74 time signals (corresponding to the 74 cells in the Tesla Model S module). The generated dataset contained 1,000 samples consisting of the simulated currents of the parallel pack. Each sample is represented as a matrix of size $74 \times t_{max}$ with $t_{max} \approx 3{,}600$ s, being the duration of the pack discharge. A total of 500 healthy and 500 faulty packs were simulated. The healthy packs are assigned parameters following the normal distribution of the model for all 74 cells, while the faulty packs contain a

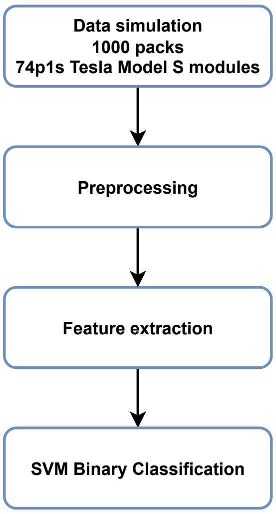


Figure 10: Data processing pipeline of the proposed method for cell fault detection.

faulty cell whose resistance $r_{faulty\ cell}$ has been set to be abnormally high (as defined at the beginning of the fault detection section). Different degrees of fault are considered during the simulations. Among the 500 simulated faulty packs, 50 are assigned the faulty cell resistance value $r_{faulty\ cell} = 1.1 \times \mu_r$, 50 are assigned the value $r_{faulty\ cell} = 1.2 \times \mu_r$, and so forth until $r_{faulty\ cell} = 2.0 \times \mu_r$.

Data Pre-processing

The first step of the data pre-processing phase involves removing a number of current sensors from the samples of the data-set. By removing the current data of the faulty cell, the performance of the fault-detection algorithm was made more robust and more representative of packs deployed in practice, as costs may limit the number of branch currents being monitored. In fact, it was found to be straightforward to distinguish a healthy pack from a faulty one when all the current distributions were measured (i.e. when all the currents flowing through the cells were known, including that of the faulty cell). By deliberately removing the data generated by the faulty cell from the algorithm's input, the fault detection problem becomes non-trivial. If the current from the faulty cell is not measured, the behaviour of the current distributions appears to be similar to that of a healthy pack, can be seen from the results of Figure 6. The method presented here thus aimed to perform faulty pack detection using only a fraction of the total potential current sensors. To do this, first, a given number of current signals, $N = N_c - N_s \geq 1$ where N_c is the total number of cells in the parallel pack and N_s is the number of sensors that are ultimately used, were removed from each sample of the dataset. This is equivalent to placing sensors on only a few of the 74 branches of the Tesla Model S's parallel pack module. For the faulty packs, the current sensor of the faulty branch is removed and the other removed current sensors are randomly selected.

The second step of the data pre-processing stage consisted in adding noise to the data and then filtering it. This noise was added to the model simulation data to make it more realistic with respect to the current sensor data measured from experiments. Given that the shunt resistors used for current sensing can generate a measurement error of approximately 0.1%,[14] a Gaussian noise with an amplitude of $\pm 0.1\%$ of the current signals was used. To be conservative, the standard deviation of the Gaussian noise distribution for each current signal was set to 0.05% of the mean value of the current during discharge. All the signals were then filtered by a low-pass Butterworth filter of order 5, and critical frequency $f_c = 0.005$ Hz (the sampling frequency being $f_s = 1$ Hz). These operations resulted in slightly distorted signals compared to the original data.

The third and last step of the data pre-processing stage is the train-test splitting of the data. The dataset was split into 80:20 proportions for the training and the testing set respectively; a 5-fold selection is applied for this purpose. These two sets contained samples consisting of matrices of size $N_s \times t_{max}$. Features are then extracted from these samples and, after the feature transformation has been applied, each sample consists of a vector of length $N_{features}$. The standard technique of training an SVM classifier[59] was then applied to the datasets.

Feature extraction

Six features, labelled as f_1 to f_6, were considered. Each sample consists of a matrix containing in each of its rows a branch current time-series. To be specific about individual time-series within those for the whole pack, the row index p is used to denote a given cell current within the parallel pack, and t will refer to the time index of the signal. The total number of cells is N_c and the duration of the time-series is t_{max}, with a time step $\Delta t = 1$ s between each sample t. A given sample of the dataset is denoted by S and the value of the current in cell p at time t within the signal is $S[p,t]$.

1. Feature 1 is the average sum of local maxima over the cell currents across the whole discharge. Writing the local current maximum on a given time interval within a cell as $\max_{t \in I} S[p,t]$, and the k$^{\text{th}}$ local maximum current encountered during its discharge (out of N local maxima) as $\left(\max_t S[p,t]\right)_k$, the feature was then defined as:
$$f_1 = \frac{1}{N_c} \sum_{p=1}^{N_c} \sum_{k=1}^{N} \left(\max_t S[p,t]\right)_k . \tag{11}$$

2. Feature 2 characterised the local current minima instead of the local maxima:
$$f_2 = \frac{1}{N_c} \sum_{p=1}^{N_c} \sum_{k=1}^{N} \left(\min_t S[p,t]\right)_k . \tag{12}$$

3. Feature 3 uses the standard deviation instead of a mean:
$$f_3 = \sqrt{\frac{1}{N_c} \sum_{p=1}^{N_c} \left[\left[\sum_{k=1}^{N} \left(\max_t S[p,t]\right)_k - f_1 \right]^2 \right]} . \tag{13}$$

4. Feature 4 was defined similarly to f_3 but used the local minima:
$$f_4 = \sqrt{\frac{1}{N_c} \sum_{p=1}^{N_c} \left[\left[\sum_{k=1}^{N} \left(\min_t S[p,t]\right)_k - f_3 \right]^2 \right]} . \tag{14}$$

5. Feature 5 was based on the standard deviation of all local maxima within the signals (rather than taking their sum as in f_3) for a given p:
$$\left\{\max_t S[p,t]\right\} = \left\{ \left(\max_t S[p,t]\right)_0 , \dots , \left(\max_t S[p,t]\right)_N \right\} .$$
We define
$$\max S = \left\{ \max_t S[1,t], \dots , \max_t S[N_c,t] \right\}$$
and denote its elements as $(\max S)_i$ which are numbered from 1 to N_m. Then, the feature, corresponding to the standard deviation of all local maxima within a sample was defined
$$f_5 = \sqrt{\frac{1}{N_m} \sum_{i=1}^{N_m} \left[\left[(\max S)_i - \overline{(\max S)_i} \right]^2 \right]} . \tag{15}$$

6. The final feature was defined as the standard deviation of all local minima within the signal S:
$$f_6 = \sqrt{\frac{1}{N_m} \sum_{i=1}^{N_m} \left[\left[(\min S)_i - \overline{(\min S)_i} \right]^2 \right]} . \tag{16}$$

SVM binary classification

The classification was performed by an SVM classifier,[60] including hyperparameter optimisation by grid search. A 5-fold cross-validation was used to create a validation set from the test set obtained during the train-test split of the pre-processing phase. The hyperparameter grid was:

- C: [0.01, 0.05, 0.1, 0.5, 1, 5, 10],
- γ: $[10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, 1]$,
- Kernel: ['*linear*', '*rbf*', '*poly*', '*sigmoid*'].

C denotes the regularisation parameter, while the kernel function transforms the input data into another, often non-linear and high-dimensional feature space, and γ determines the influence of the training samples on the definition of the SVM separation hyperplane. The parameters retained after optimisation were C = 1 and kernel = '*linear*', γ being used only for the '*rbf*', '*poly*' and '*sigmoid*' kernels.[60]

Data Availability

While no experimental data is produced, the simulation dataset generated for this study, produced by the battery pack model, can be made accessible upon reasonable request to the authors of this paper.

Code Availability

The DFN-type simulation and post-processing codes that have been used to produce the results of this study are available by the corresponding authors upon reasonable request.

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Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

PL is responsible for development of the pack model, detection methods and original article draft. RD is responsible for development of the pack model, DFN parallel model, detection methods and original article draft. ECT is responsible for development of the DFN parallel model, design of the project approach and article review. SRD is responsible for design of the project approach and article review.

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